In this question you must show all stages of your working.

Solutions based on calculator technology are not acceptable.

(i) Use integration by parts to find the exact value of

$$\int_0^4 x^2 \mathrm{e}^{2x}\,\mathrm{d}x$$

giving your answer in simplest form.

(5)

(ii) Use integration by substitution to show that

$$\int_3^{\frac{21}{2}} \frac{4x}{(2x-1)^2}\,\mathrm{d}x = a + \ln b$$

where a and b are constants to be found.

(7)